James King

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EDUCATION

Cornell University *Ithaca, NY*
B.S. Mechanical Engineering | GPA: 3.4 *Anticipated May 2028*
Relevant Coursework: Finite Element Analysis | Mechanics of Engineering Materials* | System Dynamics* | Fluid Machines* | Engineering Dynamics | Thermodynamics | Mechanical Design | Waves & Oscillations (*Fall 2026)

ENGINEERING EXPERIENCE

Knorr-Bremse (New York Air Brake) *Watertown, NY*
Mechanical Engineering Testing Intern *May 2026 – Aug 2026*

- Redesigned the lever arm of a pneumatic high-speed impact test bench from concept to build, raising lever ratio 1:4 → 1:7 and cutting weight **38% (31 → 19 lbs)** to reach a calculated **61 mph strike at 130 psi** (60 mph target).
- Sized geometry with first-principles dynamics (cylinder energy balance, rotational inertia) and verified the arm survives its own impact loads in **Ansys FEA**.
- Sourced components and produced GD&T drawings for a **$20K+ hose flex test rack** replicated at two international test sites to standardize test methodology across the company's global locations.

Cornell Racing Formula SAE *Ithaca, NY*
Ergonomics Subteam Lead / Part Designer *Nov 2024 – Present*

- Lead the subteam owning all driver-interface systems (steering, pedals, seat, harness, firewall) for ARG27; set the schedule from design review to first drive and personally design the seat, head restraint, and firewall.
- Designed and fabricated the 2026 competition steering wheel (CFRP sandwich panel, 3D-printed grips, quick-disconnect) through driver-survey-driven iterations, cutting weight **30% to 330 g**.
- Fabricated and three-point bend tested four CFRP sandwich layups to failure, varying ply schedule and core thickness; selected the 0/45/0 layup at **1,800 N (2.5× the 700 N requirement)** and the best strength-to-weight of **32.1 N/g**.
- Characterized thermal conductivity of proprietary brake pad materials, providing data that justified final material selection at design event review.

Shaw Harbor Holdings (Skip-Line LLC) *Adams, NY*
Mechanical Engineering Intern *Summers 2024 & 2025*

- Redesigned laser-cutter fixtures into drop-in locating fixtures that eliminated manual realignment between enclosure runs, **reducing machine downtime 25%**.
- Co-developed a modular enclosure platform in CAD where one base housing serves multiple product variants via swappable side panels, reducing unique part count and tariff exposure.

PROJECTS

E-Bike Frame Structural Analysis (Ansys FEA) *Cornell Course Project*

- Modeled a bike-share frame with shell elements under rider and battery loads; a **mesh convergence study** showed the initial mesh under-predicted peak von Mises stress by ~48% (29 → 56 MPa). Step-through redesign met **SF > 2**.

HertzTrap: Spotted Lanternfly Trap (MAE 2250) *People's Choice Award*

- Designed a 3D-printed field trap for client E&J Gallo using a 60 Hz acoustic lure (Arduino Nano + amplifier) and dual-polarity electrified mesh; enclosure held a **40 lb load vs. 30 lb target**, 6 lb total, redeployable in under 15 s.

Pocket AI Voice Assistant (Raspberry Pi Zero 2W) *Personal Project*

- Built an untethered voice assistant integrating local speech recognition (faster-whisper) and text-to-speech (Piper) with a cloud LLM, battery HAT, and systemd auto-start; currently optimizing end-to-end latency.

LEADERSHIP

Engineering Dean's Advisory Council (EDAC) *Sept 2025 – Present*

- Selected by Admissions as **one of 20 undergraduates** to advise the Dean of Engineering on academic policy and student experience.

TECHNICAL SKILLS

Design & Analysis: SolidWorks, NX, Creo, Inventor, Onshape | Ansys FEA | GD&T | MATLAB | Python | LabVIEW
Fabrication & Testing: CFRP Layup | Test Bench Design | Mechanical & Thermal Testing | DAQ | Machining | 3D Printing